23.10 What the Chapter Buys

Nine sections is a lot of new vocabulary, and vocabulary is not what you should carry away. This closing section introduces nothing. It names the four moves the chapter actually used, reads four earlier theorems of the book again from the height we have reached, sets out in one place which hypothesis on the field each result needed and what breaks without it, and says plainly what was quoted rather than proved and what was left outside the door on purpose.

23.10.1 Four moves

Average over the group to build an invariant object. This is the whole of Theorem 23.7.3. Given a projection onto a subrepresentation that is not invariant, replace it by the average of its conjugates $\rho(g)P\rho(g)^{-1}$ over $g \in G$; the average is invariant because translating the summation index permutes the summands, and it is still a projection onto the same subspace. The hypothesis $\operatorname{char} F \nmid |G|$ is not decoration: the average is divided by $|G|$, and if that number is 0 in F there is nothing to divide by. The same move produced the intertwiner in Theorem 23.8.7, where an arbitrary linear map is averaged into an equivariant one.

Once you have seen it, you will recognize it elsewhere: whenever a construction depends on a choice and a finite group acts on the choices, averaging over the group removes the dependence.

Use Schur's lemma to force a scalar. Theorem 23.4.1 has two halves, and the useful one in practice is the second: over an algebraically closed field, an operator commuting with an irreducible family is a scalar (Corollary 23.4.3). The pattern of use is always the same. You have an operator built by some construction, you check it commutes with everything in sight, and you conclude it is λ id; then you take a trace to find λ. That single step is what turns the averaged intertwiner of the previous paragraph into the orthogonality relations, and it is what collapses the representations of an abelian group to lines (Corollary 23.4.4).

Count with $\sum n_i^2$. Wedderburn's theorem gives $\dim A = \sum_i n_i^2$ (Corollary 23.6.2), which for a group algebra reads $|G| = \sum_i (\dim V_i)^2$ (Corollary 23.8.10). This is a very strong constraint for a small group: once you know the number of conjugacy classes, the sum of squares often has only one solution, and you have the dimensions of all the irreducible representations before you have written down a single matrix. It plays the role Rank–Nullity plays in Chapter 2: an identity that pins down what you were otherwise going to have to compute.

Read a commuting family as a commutative algebra. Chapters 9 and 12 studied families $\mathcal{F} \subseteq \mathcal{L}(V)$ whose members commute. The algebra generated by $\mathcal{F}$ is then commutative, and the chapter's theorems apply to it directly. This is not a reformulation for its own sake; it is what makes the next four paragraphs possible.

23.10.2 Four old theorems, seen from here

Commuting operators have a common eigenvector (Theorem 9.9.3). Chapter 9 proved this by restricting to an eigenspace and inducting. From here: the algebra A generated by the family is commutative, and V contains an irreducible A-invariant subspace U, namely a nonzero invariant subspace of least dimension. By Corollary 23.4.4, over an algebraically closed field U is a line, and a line invariant under every member of the family is precisely a common eigenvector. Chapter 9's hypothesis — that each characteristic polynomial splits — is what algebraic closure was standing in for. Theorem 9.9.5 then follows by decomposing V into such lines.

One orthonormal eigenbasis for a commuting normal family (Theorem 12.7.1). The same statement with the inner product carried along. The commuting normal family generates

a commutative algebra, and that algebra is closed under adjoints because the adjoint of a normal operator is a polynomial in it (Corollary 12.4.5). Closure under adjoints is exactly what makes the invariant complement available without averaging: the orthogonal complement of an invariant subspace is invariant (Proposition 23.4.6). So the family acts semisimply for free, and over $\mathbb{C}$ — where Corollary 23.4.4 applies — the decomposition into lines is orthogonal.

Circulants and the discrete Fourier transform (Theorem 12.9.4, Corollary 12.9.5, Theorem 12.9.6). Section 9 did this one in full. Chapter 12 found a single unitary $\mathbf{F}$ diagonalizing every circulant at once and could not say where it came from; it came from the character table of $\mathbb{Z}/n\mathbb{Z}$ (Theorem 23.9.6). Chapter 12 found that the product of two circulants is the circulant of the convolution of their first rows; that is multiplication in the group algebra written in the wrong basis, and in the right one it is the coordinatewise product on $\mathbb{C}^{\widehat{G}}$ (Theorem 23.9.7). Chapter 12 found that every circulant is a polynomial in the shift; that is the statement that $\mathbb{C}[\mathbb{Z}/n\mathbb{Z}]$ is generated by one element (Theorem 23.9.8). Three computations, one structure.

The commutant of a Weyr matrix (Proposition 10.4.6), **and disjoint spectra splitting a commutant** (Corollary 12.10.4). These were computations with blocks. The chapter's name for what they compute is the commutant A' of the algebra A generated by the matrix, and Theorem 23.5.5 says what to expect of it when A is semisimple: the commutant of the commutant gives nothing new. Chapter 12's disjoint-spectra result is the case where the invariant subspaces are forced apart by the spectra alone, semisimple or not, so that the answer is block diagonal whatever the blocks are like inside; Chapter 10's Weyr computation is the opposite situation, one eigenvalue and a matrix that need not be semisimple, where the double commutant theorem does not apply and the answer is correspondingly untidy — block triangular rather than block diagonal.

Quick check. Chapter 12 proved that the Fourier matrix is unitary by summing a geometric series. Section 8 proved the same relations by averaging and Schur's lemma. What does the second proof need that the first does not?

Solution. The second needs the field to be $\mathbb{C}$ — algebraically closed, so that Schur's lemma gives scalars, and of characteristic not dividing $|G|$, so that the averaging is legal — and it needs the group to be finite. The first needs only the identity $(z-1)(1+z+\cdots+z^{n-1}) = z^n-1$, and it works over any field containing n distinct n-th roots of unity, as Exercise (C3: The same theory over a finite field) shows over $\mathbb{F}_5$. The second proof explains; the first computes. Neither replaces the other.

23.10.3 Which hypothesis did what

Almost every theorem in this chapter is false over the wrong field, and the table is worth having in one place. In it F is the field of the representation and G is a finite group.

Result	Needs	What fails without it
Theorem 23.2.2	nothing	—
Theorem 23.4.1, first half	nothing	—

Result	Needs	What fails without it
Theorem 23.4.1, second half	F algebraically closed	the rotation representation of a cyclic group on $\mathbb{R}^2$ is irreducible with commutant a copy of $\mathbb{C}$
Corollary 23.4.4	F algebraically closed	the same rotation representation, irreducible of dimension 2
Corollary 23.4.5	F algebraically closed	the same rotation matrix, a commuting family of one member with no real eigenvector
Theorem 23.5.2	F algebraically closed	the same example: a proper subalgebra acting irreducibly
Theorem 23.6.1, as proved here	F algebraically closed, A semisimple	without closure, $\mathbb{C}$ as an $\mathbb{R}$-algebra is semisimple and is not a product of matrix algebras over $\mathbb{R}$; without semisimplicity, the upper triangular matrices are not one either
Theorem 23.5.5	no field hypothesis; A acts semisimply and contains id_V	for the upper triangular 2×2 matrices, A' is the scalars and $A'' = M_2(F) \neq A$
Theorem 23.7.3	$\mathrm{char}\, F \nmid \lvert G \rvert$	$\mathbb{Z}/p\mathbb{Z}$ acting on $\mathbb{F}_p^2$ by unipotent matrices: the invariant line has no invariant complement
Theorem 23.8.7	$F = \mathbb{C}$	the modular case, which the chapter does not treat
Theorem 23.9.1	$F = \mathbb{C}$	over $\mathbb{R}$ the group $\mathbb{Z}/3\mathbb{Z}$ has only the trivial character

Three comments on the table. First, Theorem 23.5.5 is the one big theorem of the chapter that needs nothing of the field at all; what it needs is semisimplicity, and that is a hypothesis about the algebra and the space it acts on. Second, the two "algebraically closed" rows about Schur and Burnside are the same failure seen twice, and the witness is the same rotation matrix both times; that is a sign the hypothesis is doing genuine work rather than covering a gap in a proof. Third, the characteristic hypothesis and the algebraic-closure hypothesis are independent. Over $\mathbb{C}$ only the first can fail, and it cannot, since $\mathrm{char}\, \mathbb{C} = 0$; that is why the complex theory is so much cleaner than the general one, and why the chapter chose $\mathbb{C}$ whenever it could.

Warning. The most tempting misreading of this chapter is "every finite-dimensional algebra over $\mathbb{C}$ is a product of matrix algebras". It is not. Theorem 23.6.1 assumes the algebra is **semisimple**, and the upper triangular 2×2 matrices over $\mathbb{C}$ are a three-dimensional algebra that is not. Acting on $\mathbb{C}^2$, that algebra leaves the line $\mathrm{span}(\mathbf{e}_1)$ invariant with no invariant complement, since $\mathbf{E}_{12}$ sends any $(x, 1)$ to $(1, 0)$; and its radical is the line spanned by $\mathbf{E}_{12}$ (Definition 23.3.5). No product of matrix algebras has a nonzero radical. Before applying

Wedderburn, check semisimplicity — and for a group algebra that check is Theorem 23.7.3, which is a statement about the characteristic and nothing else.

23.10.4 Quoted, not proved

The chapter stated six things with credit and proved none of them. Nothing in the chapter rests on any of them, and each is flagged where it appears.

- **Wedderburn–Artin over a division ring.** A finite-dimensional semisimple algebra over any field is a product of matrix algebras over division rings. We proved only the algebraically closed case, where the division rings are the field itself. The book has not built division rings, and the general theorem needs them.
- **Wedderburn's little theorem**, that a finite division ring is a field. Same reason.
- **The character table does not determine the group.** There are two nonisomorphic groups of order 8 with identical character tables. The book has neither the group theory to construct them nor the need.
- **The structure theorem for finite abelian groups**, that every one is a direct product of cyclic groups. Section 9 proved $\widehat{G} \cong G$ only for such products, and said so. Chapter 10's Smith normal form is the linear-algebra shadow of the missing argument.
- **The converse of Proposition 23.3.6**, that a finite-dimensional algebra with zero radical is semisimple. Section 3 states it and does not use it; wherever semisimplicity is needed later it is established directly.
- **When the commutant of a single matrix is the algebra of polynomials in it**, namely when the minimal and characteristic polynomials agree. Section 1 states the criterion in a remark and computes examples on both sides of it, and proves nothing about it.

23.10.5 Left outside on purpose

The subject continues, and the chapter stopped where the book's tools stopped rather than where the subject does. Four directions, in rough order of how soon you would meet them.

Induced representations and Frobenius reciprocity. Given a subgroup $H \leq G$ and a representation of H, there is a way to build a representation of G, and the characters of the two are related by an adjunction. This is the standard next tool, and it is the standard way to produce the irreducible representations of a group rather than merely count them. It needs cosets of a subgroup, which Chapter 0 does not have.

Modular representation theory. What happens when $\operatorname{char} F$ divides $|G|$ and Theorem 23.7.3 fails. Representations no longer decompose into irreducibles, the group algebra is not semisimple, and a genuinely different theory takes over. Section 7's unipotent example is the whole of what we said about it.

Character tables as a computational subject. We computed a few small ones: the symmetric group on three letters, the cyclic group $\mathbb{Z}/n\mathbb{Z}$, and the Klein four-group. Computing them in general is a craft with its own techniques, and it belongs to a course on group theory.

Representations of infinite groups. Every theorem here used finiteness twice: to average, and to count. For compact groups an integral replaces the sum and most of the chapter survives; for other infinite groups almost none of it does.

23.10.6 One more look back

The book began with one linear map and asked for a basis making it simple. Chapter 9 found eigenvectors, Chapter 10 found the Jordan and rational forms, Chapters 11 to 13 found orthonormal bases when there was an inner product to respect. Then the question widened: Chapter 9 Section 9 and Chapter 12 Section 7 asked for a basis good for a whole commuting family at once. This chapter widened it once more, to a set of operators closed under composition, and found that over an algebraically closed field there is essentially one building block, a full matrix algebra, and that everything semisimple is a product of those.

That is a satisfying place to stop, but notice what it cost. The answer is clean only over an algebraically closed field, only for semisimple algebras, and — on the group side — only when the characteristic is coprime to the order. Every one of those three conditions was a real restriction with a two-by-two counterexample behind it. A structure theorem that applies to everything is usually a structure theorem that says nothing; this one says a great deal, and the price is printed on the label.

23.10.7 Exercises

23.10.7.1 A. Check your understanding

Exercise. A1

(a) Name the four moves of this chapter and, for each, one theorem in which it was used.

(b) State the field hypothesis needed by each of Theorem 23.4.1 (second half), Theorem 23.5.2, Theorem 23.6.1, and Theorem 23.7.3.

(c) Determine whether the following are true, with a reason. (i) Every finite-dimensional complex algebra is a product of matrix algebras. (ii) Every complex representation of a finite group decomposes into irreducibles. (iii) Every representation of a finite group over every field decomposes into irreducibles. (iv) The book proves that two nonisomorphic groups can share a character table.

(d) Why is $|G| = \sum_i (\dim V_i)^2$ useful before any matrix has been written down?

Solution.

(a) Averaging over the group, used in Theorem 23.7.3 and Theorem 23.8.7; forcing a scalar with Schur, used in Corollary 23.4.3 and Corollary 23.4.4; counting with $\sum n_i^2$, used in Corollary 23.6.2 and Corollary 23.8.10; reading a commuting family as a commutative algebra, used throughout Section 9 and in the rereading of Theorem 9.9.5 above.

(b) Schur's second half and Burnside both need F algebraically closed. Theorem 23.6.1, in the form proved here, needs F algebraically closed and the algebra semisimple. Theorem 23.7.3 needs $\operatorname{char} F \nmid |G|$ and nothing else.

(c) (i) False: the upper triangular 2×2 matrices are a counterexample, since they are not semisimple. (ii) True, by Corollary 23.7.4, since $\operatorname{char} \mathbb{C} = 0$ never divides $|G|$. (iii) False: $\mathbb{Z}/p\mathbb{Z}$ on $\mathbb{F}_p^2$ by unipotent matrices. (iv) False. The statement is quoted with credit and not proved.

(d) Because the number of summands is the number of conjugacy classes (Theorem 23.8.9), and for a small group the equation $|G| = \sum n_i^2$ with that many positive integer summands, one of which is 1 for the trivial representation, usually has a unique solution.

The dimensions come out of arithmetic rather than out of linear algebra.

23.10.7.2 B. Practice

Exercise. B1: Two group algebras of order six

(a) Write down the Wedderburn decomposition of $\mathbb{C}[\mathbb{Z}/6\mathbb{Z}]$, with a reason for each factor.

(b) The group S_3 has three conjugacy classes. Use Corollary 23.8.10 to find the dimensions of its irreducible complex representations, and write down the Wedderburn decomposition of $\mathbb{C}[S_3]$.

(c) Check both answers against $\dim \mathbb{C}[G] = |G|$.

Solution.

(a) $\mathbb{Z}/6\mathbb{Z}$ is abelian, so all irreducible representations are one-dimensional (Theorem 23.9.1) and there are six of them (Proposition 23.9.3). Hence

$$\mathbb{C}[\mathbb{Z}/6\mathbb{Z}] \cong \mathbb{C} \times \mathbb{C} \times \mathbb{C} \times \mathbb{C} \times \mathbb{C} \times \mathbb{C},$$

which is Theorem 23.9.7 for this group.

(b) Three conjugacy classes means three irreducible representations (Theorem 23.8.9), of dimensions $n_1, n_2, n_3 \geq 1$ with $n_1^2 + n_2^2 + n_3^2 = 6$ by Corollary 23.8.10. The only solution in positive integers, up to order, is $1 + 1 + 4$, so the dimensions are $1, 1, 2$. Therefore

$$\mathbb{C}[S_3] \cong \mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C}).$$

(c) In (a), $6 \cdot 1 = 6$. In (b), $1 + 1 + 4 = 6$. Both match $\dim \mathbb{C}[G] = |G|$, which is the dimension count of Corollary 23.6.2.

Exercise. B2: What the sum of squares allows at order eight

Let G be a group of order 8.

(a) List all the possible multisets of dimensions of the irreducible complex representations of G. *Remember that the trivial representation is always present.*

(b) Say which of your lists corresponds to an abelian G, and why.

(c) Deduce how many conjugacy classes a nonabelian group of order 8 has.

Solution.

(a) By Corollary 23.8.10 the dimensions satisfy $\sum n_i^2 = 8$ with every $n_i \geq 1$. Let k be the number of n_i equal to 2 or more. Since $3^2 = 9 > 8$, every n_i is 1 or 2, so $8 = 4k + (\text{number of ones})$. For $k = 0$ we get eight 1s; for $k = 1$, four 1s and one 2; for $k = 2$ we would get no 1s at all, which is impossible because the trivial representation has dimension 1. So the two possibilities are $(1, 1, 1, 1, 1, 1, 1, 1)$ and $(1, 1, 1, 1, 2)$.

(b) The first. By Theorem 23.9.1 an abelian G has all irreducible representations one-dimensional, and conversely if all dimensions are 1 then by Theorem 23.6.1 the algebra $\mathbb{C}[G]$ is a product of copies of $\mathbb{C}$, hence commutative; since the elements δ_g commute

in $\mathbb{C}[G]$, the group is abelian.

(c) A nonabelian G of order 8 has dimension list $(1,1,1,1,2)$, so five irreducible representations, and by Theorem 23.8.9 five conjugacy classes.

23.10.7.3 C. Going deeper

Exercise. C1: The circulants as a semisimple algebra

Let $n \geq 2$ and let $\mathcal{C}_n \subseteq M_n(\mathbb{C})$ be the algebra of circulants of Chapter 12 Section 9.

(a) Explain why $\mathcal{C}_n$ acts semisimply on $\mathbb{C}^n$, and write down its Wedderburn form as an abstract algebra.

(b) Prove that an $\mathbf{X} \in M_n(\mathbb{C})$ commutes with every circulant if and only if $\mathbf{F}^*\mathbf{X}\mathbf{F}$ is diagonal, and deduce $\mathcal{C}_n' = \mathcal{C}_n$.

(c) Verify Theorem 23.5.5 for $A = \mathcal{C}_n$.

Hint for (b): first show that a matrix commuting with every diagonal matrix is diagonal, by testing against the matrix units $\mathbf{E}_{kk}$.

Solution.

(a) First, $\mathcal{C}_n$ acts semisimply on $\mathbb{C}^n$. By Theorem 12.9.1 (b) the adjoint of a circulant is a circulant, so if U is $\mathcal{C}_n$-invariant and $\mathbf{x} \in U^\perp$, then for every $\mathbf{C} \in \mathcal{C}_n$ and $\mathbf{u} \in U$ we have $\langle \mathbf{C}\mathbf{x}, \mathbf{u}\rangle = \langle \mathbf{x}, \mathbf{C}^*\mathbf{u}\rangle = 0$, because $\mathbf{C}^*\mathbf{u} \in U$; so $U^\perp$ is an invariant complement. Second, as an abstract algebra $\mathcal{C}_n \cong \mathbb{C}[\mathbb{Z}/n\mathbb{Z}]$ by Theorem 23.9.8, and by Theorem 23.9.7 its Wedderburn form is the product of n copies of $\mathbb{C}$; all the blocks of Theorem 23.6.1 are 1×1 because the algebra is commutative.

(b) First the hint. Suppose $\mathbf{Y} \in M_n(\mathbb{C})$ commutes with every diagonal matrix. Taking $\mathbf{D} = \mathbf{E}_{kk}$, the matrix $\mathbf{Y}\mathbf{E}_{kk}$ has k-th column equal to the k-th column of $\mathbf{Y}$ and zeros elsewhere, while $\mathbf{E}_{kk}\mathbf{Y}$ has k-th row equal to the k-th row of $\mathbf{Y}$ and zeros elsewhere. Equating them forces $y_{jk} = 0$ for $j \neq k$ and $y_{kj} = 0$ for $j \neq k$. Letting k run, $\mathbf{Y}$ is diagonal.
Now let $\mathbf{X}$ commute with every circulant and put $\mathbf{Y} = \mathbf{F}^*\mathbf{X}\mathbf{F}$. By Corollary 12.9.5 (c) the circulants are exactly the matrices $\mathbf{F}\mathbf{D}\mathbf{F}^*$ with $\mathbf{D}$ diagonal, and

$$\mathbf{X}(\mathbf{F}\mathbf{D}\mathbf{F}^*) = (\mathbf{F}\mathbf{D}\mathbf{F}^*)\mathbf{X} \iff \mathbf{Y}\mathbf{D} = \mathbf{D}\mathbf{Y},$$

after multiplying by $\mathbf{F}^*$ on the left and $\mathbf{F}$ on the right and using $\mathbf{F}^*\mathbf{F} = \mathbf{I}$ (Theorem 12.9.3). So $\mathbf{X}$ commutes with every circulant exactly when $\mathbf{Y}$ commutes with every diagonal matrix, that is exactly when $\mathbf{Y}$ is diagonal, that is exactly when $\mathbf{X} \in \mathcal{C}_n$. Hence $\mathcal{C}_n' = \mathcal{C}_n$.

(c) By (b), $\mathcal{C}_n'' = (\mathcal{C}_n')' = \mathcal{C}_n' = \mathcal{C}_n$. And $\mathcal{C}_n$ acts semisimply on $\mathbb{C}^n$ by (a) and contains $\mathbf{I}$, so Theorem 23.5.5 applies and predicts exactly this. The example also shows the theorem has content: for the algebra of scalar matrices, which is also semisimple and contains $\mathbf{I}$, the commutant is all of $M_n(\mathbb{C})$ and the double commutant is the scalars again — the same conclusion reached along a very different route.